

Variety-specific sugarcane yield simulations and climate change impacts on sugarcane yield using DSSAT-CSM-CANEGRO model

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ABSTRACT

Crop simulation models are still little used for sugarcane crops due to the lack of understanding of their capabilities and experience in calibration as compared to other crops. Realistic assessment of future environmental change effects on crop production is also necessary for successful agricultural management and planning. The objective of this study is to provide DSSAT-CANEGRO simulated variety-wise sugarcane yield models for twelve different locations of Muzaffarnagar District of India. The CANEGRO model calibration is performed under three different dates of planting (early, mid, late) during the spring season 2013–2014. The results of the models are validated for consecutive two years datasets (2014–15 and 2015–16). The model shows the best calibration and validation results under the mid planting date scenario (simulated yield +0.85 %, +2.80 % and +5.20 %). The impact of climate change (sensitivity analysis) on yield of sugarcane has also been made putting different values of Tmax (± 1 to ± 3 °C), Tmin (± 1 to ± 3 °C), solar radiation (± 1 to ± 3 MJ/m²/day), and atmospheric CO₂ concentration of 380 ppm (720 ppm scenario A2, 500 ppm scenario B2). The study highlighted that Sugarcane yield simulation mid-planting model presenting the highest R (0.81, 0.83), and D (0.88, 0.91), and the lowest errors (RMSE = 8.37 q ha⁻¹, 10.70 q ha⁻¹ and MAPE = 1.10 %, 1.30 %) for the years 2014–15 and 2015–16 as compared to the other two models. The CANEGRO model simulated yield under incremental values of Tmax and Tmin in the range of (+1 to +3 °C) shows gradual decrement in the yield ranges while, gradual increment of solar radiation from 1 to 3 MJ/m²/day showed a yield increment.

1. Introduction

Agriculture plays a critical role in the economy of numerous nations, particularly in developing countries (Virnodkar et al., 2020). Crop growth variability is related to various parameters that can be time dependent (rainfall, temperature, relative humidity, sunshine hours, etc.) or time-independent (previous land use, soil type, soil depth, topography, etc.) (Bégué et al., 2008). Crop modelling potentially can contribute to nutrition and global food security (Mehrabi and Sepaskhah, 2020). Crop simulation models are computer-based software application tools which are used to evaluate the effect of agronomic, hydrologic, edaphic and climatic factors on crop growth and yield. These models are quantitative tools based on scientific knowledge that conceptually present a simplified crop production system. Crop simulation models have been developed by understanding the interaction of all underlying environmental conditions and plant growth processes as

well as factors of weather, soil, pests, irrigation, and other management practices that affect plant growth and yield (Battisti et al., 2018). Therefore, they can be used for local and global applications under different management conditions, which may not be possible through empirical or regression modelling. The general indication with respect to the use of simulation models in crop yield forecasting is that the models hold promise as a tool for the future. The accuracy of the method depends partly on the adequacy of the crop model and partly on the technique used to complete the seasonal data record (Duchon, 1986). Model outputs should be verified during the development phase of a forecasting system using accurate data from the area.

The crop yield simulation model is an important tool for planners, scientists, and farm managers. It can predict yield under different soil and climatic conditions. These models simulate the development and growth of a crop through its different growth stages until maturity (Hammer et al., 1996b). Input data required by simulation models

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commonly relate to climate variables, properties of soil, genetic coefficients, and field management factors. Climate data usually include daily solar radiation, rainfall, minimum and maximum temperatures (Duchon, 1986; Lourens, 1995). The advancement of simulation models begun after the accomplishment of a model created by Ritchie (1972) to anticipate evapotranspiration from the row crops. The model is more empirical and as an input, it required seasonal variation in Leaf Area Index (LAI). Several simulation models viz. DSSAT (Hoogenboom et al., 1999), WOFOST (Boogaard et al., 1998), CROPSYST (Stockle et al., 1994) and INFOCROP (Aggarwal et al., 2006) have been used in recent years to anticipate regional yield forecast for various crops. India is the second largest producer of sugarcane (Sonkar et al., 2020). A few studies have been reported using the simulation crop growth model in India. Sugarcane is an important crop in the tropical and sub-tropical regions due to its high sucrose content (Gilbert et al., 2008). It is also the world's largest biomass crop, which is commercially produced in over 100 countries (White et al., 2020). This crop is a good source for the production of sugar, ethanol and electricity (De Aquino et al., 2017). There is a need to assess the efficiency capability of sugarcane in various agro-climatic zones of the nation, therefore, various sugarcane models, such as APSIM, CANEGRO, and QCANE have been introduced. The APSIM, CANEGRO, and QCANE are three main models used for sugarcane simulation. The APSIM and CANEGRO are currently used in the major sugarcane producing regions of the world, whereas QCANE is a less used well-known model. The APSIM and QCANE are Australian models, while CANEGRO is a South African model (O'Leary, 1999). The CANEGRO sugarcane model employs the DSSAT that has been widely utilized in America, Africa, and Asia (O'Leary, 2000).

In recent years, several yield forecasting studies have been conducted for sugarcane crops using the DSSAT-CANEGRO simulation model. Singh et al. (2010) calibrated and validated the CANEGRO model for four different field locations (Lucknow, Kushinagar, Gorakhpur, and Varanasi) in Eastern Uttar Pradesh region of India. This study revealed that the CANEGRO model satisfactorily simulated the potential development and yield. In this model, stalk fresh mass, stalk height, and sucrose yield within $\pm 15\%$ of range as compared to the observed values.

Brazil is the largest sugarcane producer country in the world and sugarcane is the most important bioenergy crop, as well as the main source of sugar (Cardozo et al., 2015). Marin et al. (2011) parameterized and assessed the CANEGRO model for Brazilian sugarcane. This model is parameterized for two sugarcane varieties at four different locations in Southern Brazil. In each sugarcane variety, ten genetic coefficients out of 20 were optimized utilizing the leave-one-out cross-validation technique of Generalized Likelihood Uncertainty Estimation (Mertens et al., 2004). The model forecasts are assessed with the assistance of soil-water content, stalk, sucrose content, aerial dry mass, and LAI. The quality of forecasts is figured out utilizing the bias, model efficiency, root mean squared error, correlation coefficient, and agreement index. The CANEGRO model has also provided satisfactory results for sugarcane modelling in southern Brazil (Nassif et al., 2012). Dias and Sentelhas (2017) evaluated the performance of three crop yield forecast simulation models (APSIM, FAO-AZM, and DSSAT-CANEGRO), in a single and multi-model approach for the commercial managed sugarcane fields of Brazil. The weather data Tmax, Tmin, rainfall, sunshine hours, and solar radiation are used in the model. Yield data from several seasons (2000-01-2013-14) and sixteen locations are utilized for model calibration and validation. The three models presented a weak performance, with low precision (R^2) and high mean absolute error (MAE). The multi-model approach, by using the ensemble of APSIM, FAO-AZM, and DSSAT-CANEGRO, improved the sugarcane yield estimates as well as minimizing the errors.

Bhengra et al. (2016) calibrated and validated the CANEGRO sugarcane model with the help of field experiment data from 1997 to 2013 in the northern part of India (four locations). The 43 years of weather parameters (minimum and maximum temperature, rainfall, and solar radiation) from 1971 to 2013, and other input parameters, such as soil

property, nitrogen content, dates of planting, maturity, harvesting, and cane yield information are collected. The twenty genetic coefficients for ten sugarcane varieties are assessed by repeated interaction in the model counts until a nearby match among observed and predicted phenology, growth, and yield is achieved. Parmar et al. (2019) calibrated and validated the CANEGRO model for the south Gujarat region, India. They utilized daily weather data from 2012-13 to 2016-17 and estimated the genetic coefficient for sugarcane variety (Co-99004 and Co-86032) by using the GLUE program (inbuilt package of DSSAT software).

Poor model estimates may not entirely be due to the functional relations that are involved in the model but may also be due to errors associated with input parameters. The successful application of any crop growth model for Indian conditions should invariably consider both these aspects because of varied agro-climatic and physiographic conditions. Therefore, a sensitivity analysis for some input parameters were carried out in various studies. Marin et al. (2013) and Singels et al. (2014) used sugarcane models to evaluate sugarcane yield responses to future climate situations. Ramachandran et al. (2017), has utilized the CANEGRO-SUGARCANE, CERES-RICE and CROPGRO legume modules to simulate the future yield under varying climate scenarios for the Chengalpet District of Tamilnadu, South India. The RCP 4.5 (representative concentration pathways) was chosen for simulation, which is good coherence with the climate of the Indian subcontinent (Chaturvedi et al., 2012). Abbas et al. (2020) studied the effects of climate on rice yield in Punjab, Pakistan. The Impact of temperature and rainfall on rice production in Punjab is also investigated by Abbas and Mayo (2021). Dias et al. (2016) have identified the rice growth (varieties At362 and Bg357) and yield variation under changing climatic conditions in the Nilwara river basin of Sri Lankans using DSSAT model. The study reveals that the increasing temperature and solar radiation and decreasing precipitation influences both growth and yield of rice. Jones et al. (2014) have evaluated the DSSAT-CANEGRO sugarcane model for simulating climate change impacts (maximum temperature, minimum temperature, rainfall and atmospheric CO₂ concentration) at 10 different experimental sites in seven countries (Brazil, South Africa, Zimbabwe, Ecuador, Reunion and USA). DSSAT-CANEGRO version 4.5 model (Singels et al., 2008) is also adjusted with improved ability for simulating high temperature, water shortage and CO₂ impacts on crop development and growth (Singels et al., 2013). Knox et al. (2010), have assessed the spatiotemporal effects of environmental change impacts on irrigation water requirements and yield for sugarcane crop growth in Swaziland. They combine the outputs of the HadCM3 (general circulation model), DSSAT-CANEGRO (sugarcane growth model) and GIS.

In India, industrialization is increasing, lifestyle is changing and the use of fossil fuel is increasing therefore CO₂ concentration in the atmosphere is increasing. From 2005–2008, India experienced dramatic growth in fossil-fuel consumption and CO₂ emission at 5.7 % per year and became the world largest fossil-fuel CO₂ emitting country (Andres et al., 2009). The CANEGRO simulated yields were correlated in view of the expected future scenario of climate change in northern Uttar Pradesh, as per the report of the Intergovernmental Panel on Climate Change (IPCC-2014). The IPCC has developed four possible CO₂ emission scenarios viz., A1, A2, B1, and B2 for every country, according to its topography, economical, technological and social backgrounds. The A2 (high emission) and B2 (low emission) scenario were taken under this study. For the A2 and B2 scenario, CO₂ was fixed at 720 ppm & 500 ppm, respectively (Arnell, 2004). These values of CO₂ had been close to the conditions under which sugarcane has been examined for CO₂ effects (De souza et al., 2008; Vu and Allen, 2009). The baseline value was simulated at 380 ppm of CO₂.

The aim of this study is to provide DSSAT-CANEGRO simulated variety-wise crop yield models for sugarcane crop. The sensitivity analysis (impact of various climatic parameters) of these generated models is also taken into consideration using the various experimental values of Tmax, Tmin, solar radiation and CO₂ concentration on sugarcane yield. Simulation modelling is one of the ways that integrate all

factors (such as, various climatic parameters, soil type, cultivar type, different dates of planting, field preparation, irrigation, control of pests and disease, fertilizers, etc.) that affects crop yield, therefore, DSSAT-CANEGRO simulation model is used to predict sugarcane yield for various locations in Muzaffarnagar District of India.

2. Study area

This study was carried out in twelve different locations of Muzaffarnagar District of Uttar Pradesh state, India, which geographically extends between Latitude $29^{\circ} 14' 28.35''$ N - $29^{\circ} 42' 36.68''$ N and Longitude $77^{\circ} 03' 45.26''$ E - $78^{\circ} 11' 35.04''$ E. The district boundary is extracted with the help of the Survey of India (SOI) toposheets (53/G and 53/K, scale 1:250,000). According to the National Informatics Centre (NIC), Muzaffarnagar, the total geographical area of the district was about 4049 km² in the year 2010–11. To understand the effects of climate variability on sugarcane yield, weather, soil, variety and agronomic management data were used in conjunction with a CANEGRO model for Chhapar sugarcane farm locations (Latitude/Longitude:

29.56° N / 77.76° E). This site is a flat terrain, an average elevation of 242 m above sea level, and falling into the Upper Gangetic Plain Region. The major portion of an area consists of common three different types of soils such as, Sandy Loam, Loam and Clay Loam (SOI report, 2003). The study area is shown in Fig. 1.

Muzaffarnagar district lies in the hot sub-humid Agro Ecological Sub Region (AESR). On the basis of rainfall, terrain, and soil characteristics, the agro-climatic zone of this region is the Western Plain Zone (Kumar, 2005). There are three weather conditions viz. summer, winter, and rainy seasons. The summer is hot and the average, most extreme temperature varies between 39° C and 45° C. The normal yearly precipitation ranges between 627 and 1130 mm, 80 % of which is received during the peak monsoon in August and September. The elevation ranges from 260 m in the North-East to 230 m in the South-West (Umar et al., 2009). The drainage pattern is entirely governed by the two major rivers (Yamuna and Ganga). The tributaries of the Yamuna river are Hindon, Krisni, Hari river, and the Katna Nala, while the major tributary of Ganga is the Solani river. This region has excellent and multiple ways of irrigation such as dug wells, tube wells, and canals. About 80 % of the

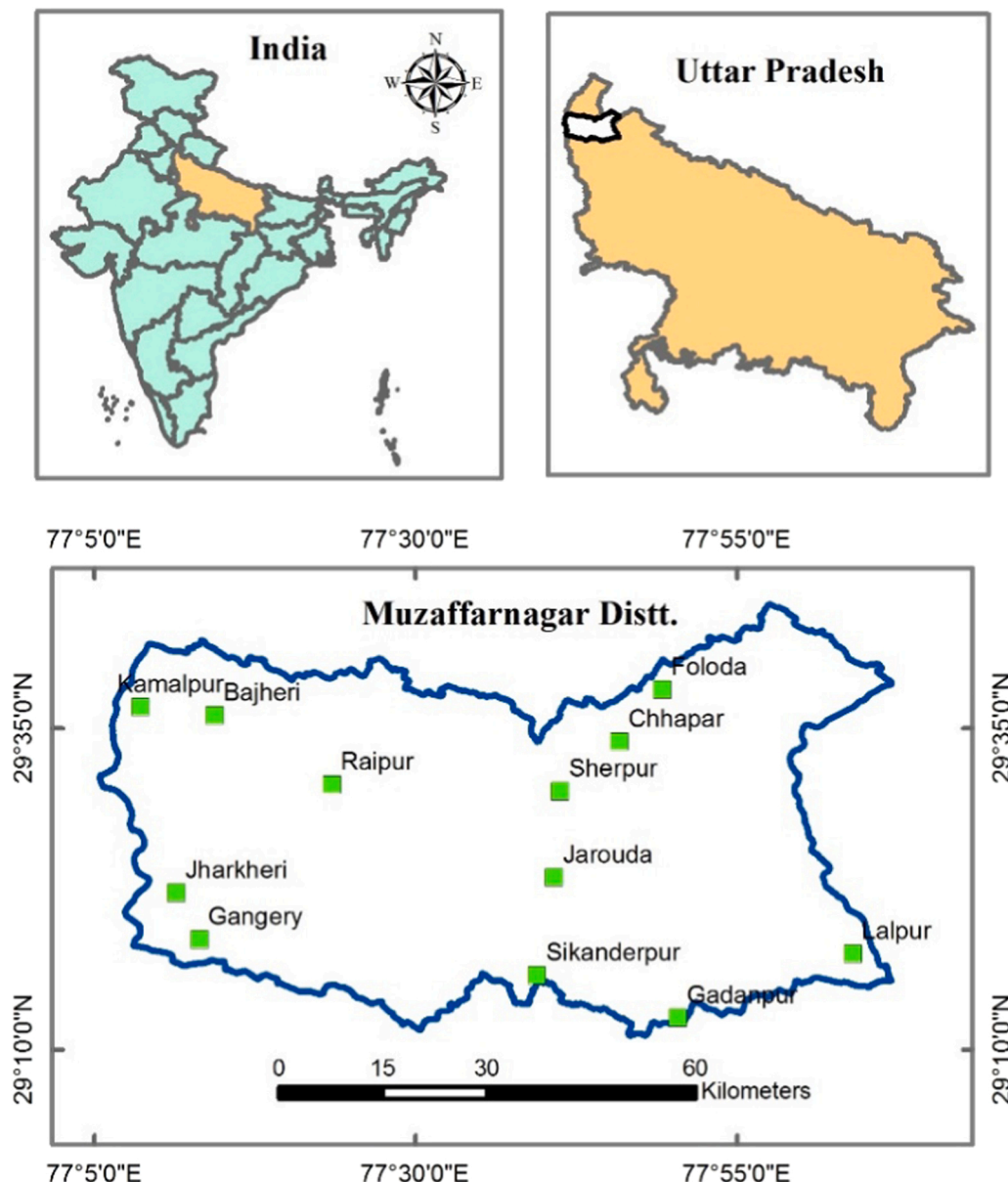


Fig. 1. Sugarcane crop field locations in Muzaffarnagar district (UP).

total geographical area of this district is cultivated sugarcane as one of the major commercial crops occupying 45–75 % of the sown area, while the wheat and paddy are the other important crops of this region (Sankhyakiya Patrika, 2014).

3. Materials and experimental methods

In DSSAT software, the CANEGRO model has been used for sugarcane yield estimation at twelve well-distributed locations of Muzaffarnagar District. The three planting dates (early, mid, late) for sugarcane crops are taken to perform various experiments with different soil conditions and different farming practices to simulate the results and to propose appropriate solutions. The main experimental file is generated in DSSAT software, which consists of all other files, such as weather file, soil file, genotype file, crop management file, input and output file, and simulation control, etc. The various soil parameters, crop management factors (field preparation, crop sowing to harvesting details, number of irrigations), and other ancillary information were collected during field visits in the year 2013–14. The one year (2013–14) dataset was used for model calibration. While the remaining two consecutive years (2014–15 and 2015–16) datasets were used for model validation. A total of 180 treatment combinations (for 12 different locations, five sugarcane varieties, and three planting dates) were used for sugarcane yield simulation modelling. The flow chart of the methodology is presented in Fig. 2.

3.1. The DSSAT Model

DSSAT is a crop simulation model, which is assembled and distributed by the International Consortium for Agricultural System Analysis (ICASA). It is a powerful crop modelling system for assisting agronomic decision-making (Hoogenboom et al., 1999; Jones et al., 2003). It simulates the effect of soil moisture, weather condition, genotype, soil, and crop nitrogen elements in plant development rather than on grain yield

(Jones et al., 2003). It additionally reproduces the impact of CO₂ on photosynthesis, which is useful in climate change impact studies. The models which are incorporated in DSSAT include CERES model for cereal crops, such as rice, wheat, maize, sorghum, millet, and barley; the CROPGROW model for soybean, peanut, and phaseolus bean; SUBSTOR model for root crops such as potato and cassava; CANEGRO for sugarcane and CROPSIM for other crops (Tsuji et al., 1994; Jones et al., 2003). In this research work, the DSSAT-CANEGRO model is used for sugarcane yield estimation study and climate change impacts on yield.

3.2. CANEGRO model description

This model was developed at the South African Sugarcane Research Institute (SASRI) in the 1980s. The CANEGRO draws its lineage from the CERES-Maize and CropGro models. It considered a valuable research platform as well as a useful tool for sugarcane modelling applications. It analyses the effects of age and season on sugarcane growth, sucrose accumulation, production potential assessment and benchmarking, crop forecasting, and investigating agronomic practices for bioenergy production (Inman-Bamber, 1991; Inman-Bamber et al., 1998; McGlinchey, 1999; Cheeru-Nayamuth et al., 2000). A new modular system Cropping System Model (CSM) and many utilities were introduced in DSSAT version 4, which offers numerous advantages to compatible crop sub-models (Plant Modules) (Jones et al., 2003). After replacing an earlier CANEGRO version 3.5, a new CANEGRO version (Singels et al., 2008) has been incorporated with the DSSAT 4.5 version (Jones et al., 2003; Hoogenboom et al., 2010). The CANEGRO model simulates sugarcane crop growth and development, including phenology, canopy development, tillering, biomass accumulation and partitioning, root growth, water stress, and lodging (Inman-Bamber, 1991; Singels and Bezuidenhout, 2002). It uses various parameters of soil that regulate the soil-water balance, cultivar, irrigation, other management information, and daily weather data (solar radiation, maximum temperature, minimum temperature, rainfall). Wind velocity and relative humidity are not

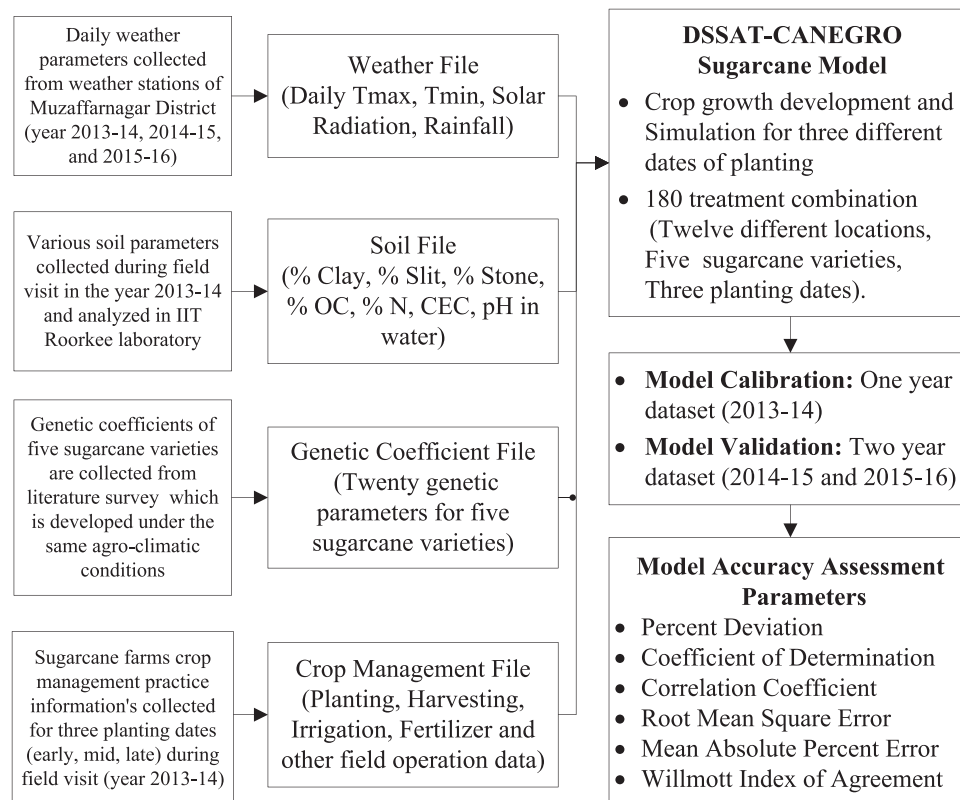


Fig. 2. Approach to simulation modelling for sugarcane crop.

essential, but rather are prescribed wherever available. In this research work, DSSAT version 4.6 software was used for sugarcane yield modelling. It was provided by Professor Gerrit Hoogenboom, University of Georgia, USA, and over forty-two crop models have been developed in this version.

3.3. Various file generation in DSSAT model

The DSSAT software was designed to perform “Experiments” to simulate the results of the complicate interactions among different farming practices, different climate, different soil conditions and to propose appropriate solutions for site-specific issues (Basak et al., 2010). Various generated files and flow of data in the DSSAT model are shown in Fig. 3. In this study, the main experiment directory file (EXP.DIR) is generated. This file consists of all the input and output files, such as weather file, soil file, genotype information file, crop management file, and simulation control, etc.

3.4. Weather file

In the weatherman module, the weather file (WTH.DIR) consists of the latitude/longitude information of weather stations, daily maximum temperature, minimum temperature, air temperature (°C), solar radiation (MJ/m²/day), and rainfall (mm). WGEN sub-module of weatherman generates daily data from monthly or weekly data. The DSSAT uses weather parameters in the sequence of Julian date so all data are compiled in the required manner. The new weather file for the Muzaffarnagar District was created for the years 2013–14, 2014–15, and 2015–16. The daily weather parameters were collected from the weather station of the Sugarcane Development Centre, Muzaffarnagar.

3.5. Various soil parameters analysis

The soil directory file (SOL.DIR) consists of various physical-chemical properties of soil, such as texture information of soil (percent sand, silt and clay), pH in water, percent of organic carbon (OC), percent of nitrogen (N), cation exchange capacity (CEC), etc. The soil sample data from the soil profile extending up to 120 cm depth were collected during the field visit from 2013 and 2014, and their soil parameters were analyzed in the soil testing laboratory of Indian Institute of Technology Roorkee. At each field location, 350 gm soil sample has been taken for five replicate positions along the vertical profile (depth-bottom 0–5, 5–30, 30–60, 60–90 and 90–120 cm). Total twelve different locations were selected in sugarcane crop fields in the district, namely; Fodda, Chhapar, Bhajeri, Sherpur, Jharkheri, Raipur, Gangery, Jarouda, Sikanderpur, Gadanpur, Lalpur, and Kamalpur. The locations selected for this study fairly cover the entire area of the district. A total of 180 soil samples were collected from sugarcane fields.

In this study, soil particle size was determined by utilizing mechanical sieve analysis and hydrometer analysis, as suggested by the American Society for Testing and Materials (ASTM). Soil samples from twelve different locations and five replicate positions along the vertical profile have been taken for grain size analysis. The extracted samples oven-dried for 24 h at 105 °C after that the required amount of sample was weighed for sieve and hydrometer analysis. Particularly, for the hydrometer analysis, 50 g of the oven-dried sample that passes through the 200 (75 µm) sieve size was taken. The hydrometer analysis is applied to separate finer soil particles (silt and clay) that work on the principle of Stoke’s Law. In six cases, the textural classification of the experimental field soil falls in the sandy loam portion of the textural triangle, while in three cases, the soil is falling in the loam and silty loam portion.

The pH of the soil is determined by mixing air-dried soil with a neutral salt solution, such as CaCl₂. The pH of soil samples was examined in the suspension of soil-water (1:5) with the assistance of a photo volt, pH meter utilizing a glass electrode. The organic carbons (OC) were examined by Walkley and Blacks rapid titration technique (Walkley and Black, 1934; Enang et al., 2018). This technique is based on the oxidation of organic matter by a mixture of K₂Cr₂O₇ (potassium dichromate) and H₂SO₄ (sulfuric acid). Unused K₂Cr₂O₇ is back titration with Fe (NH₄)₂(SO₄)₂ (ferrous ammonium sulfate). In this research work, a semi-micro Kjeldahi digestion method is used to estimate the total nitrogen status in soil (Kjeldahl, 1883; Sáez-Plaza et al., 2013). The soil samples are digested with cone, sulphuric acid to convert organic nitrogen to NH₄⁺-N. The amount of NH₄⁺-N in the digest can be determined by colorimetric methods. The soil cation exchange capacity (CEC) is determined by using the following formula:

$$\text{CEC, meq} / 100\text{g} = \frac{C * 10 * 10^{-3} * 100}{W_s * 14} \quad (1)$$

Where, C=measured concentration of NH₄⁺-N, in the percolate, obtained from the standard curve in micrograms per millilitres (µg/ml), and W_s = weight of air-dried sample used in grams (gm).

3.6. Soil profile creation

The soil module (S build) was used to generate soil files into the DSSAT model. Various soil parameter information, such as analysis date, determination methods, soil depth information, pH in water, organic carbon, cation exchange capacity, nitrogen, etc., are used to create soil files. All the computed and collected soil parameters from laboratory experiments or field visits were entered into the five replicate positions along with the vertical profile of soil (depth-bottom 0–5, 5–30, 30–60, 60–90 and 90–120 cm). Soil profiles are created for twelve different experimental locations of the entire Muzaffarnagar District. The generated twelve soil files and various soil parameters are shown in Table 1.

3.7. Genetic coefficient file (cultivar file)

The cultivar (CUL.DIR) file includes the genetic coefficient (quantified traits) which distinguish one cultivar from another in terms of crop development and growth. Genetic coefficients can be generated using AT create module of DSSAT. The genetic coefficient of any cultivar whose growth pattern resembles with the selected cultivar, which pre-exists within the model, was selected for modification running the DSSAT model using actual soil, weather, and crop management data and made it usable by adjusting various coefficients through an iterative process until the observed and simulate crop parameter data matched.

In the present study, we have used the genetic coefficient of five sugarcane varieties (CoS-767, CoSe-95422, CoS-8436, CoSe-92423, and CoSe-98231) which covers 70 %–80 % sugarcane area in Muzaffarnagar District. All these varieties are used for experimental purposes. The genetic coefficients of these varieties were developed under the same agro-climatic conditions (Varanasi, Kushinagar, Pantnagar, Gorakhpur, and

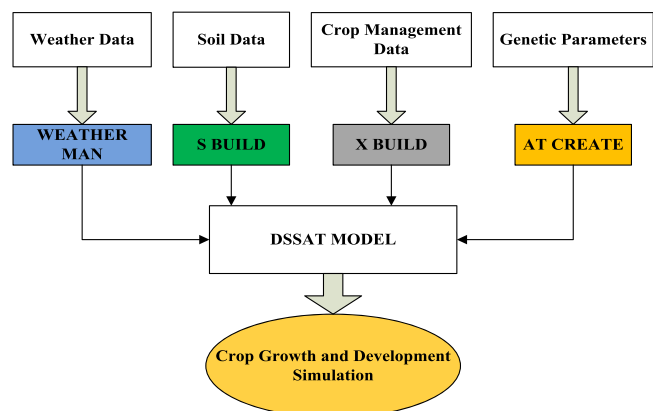


Fig. 3. Flow of data in the DSSAT model.

Table 1
Soil parameters at twelve different locations of Muzaffarnagar district.

Site Location	Soil Depth (cm)	Clay (%)	Silt (%)	Stone (%)	Organic Carbon (%)	pH in water	Cation exchange capacity, (cmol)/kg	Total Nitrogen (%)
Foloda location 1 (Latitude/Longitude: 29.634 N/ 77.819 E, Texture: Loam)	0–5	13	45	0	0.54	7.3	14.4	0.03
	5–30	11	42	0	0.54	7.3	14.4	0.03
	30–60	12	43	0	0.53	7.3	14.3	0.03
	60–90	16	39	1	0.54	7.3	14.1	0.02
	90–120	15	41	0	0.54	7.2	14.1	0.02
Chhappar location 2 (Latitude/Longitude: 29.566 N/77.764 E, Texture: Sandy Loam)	0–5	10	36	0	0.27	7.6	19.1	0.02
	5–30	11	35	0	0.27	7.6	19.1	0.02
	30–60	14	35	0	0.26	7.6	18.8	0.03
	60–90	7	29	0	0.26	7.5	18.7	0.02
	90–120	8	30	0	0.26	7.6	18.4	0.02
Bajheri location 3 (Latitude/Longitude: 29.600 N/ 77.240 E, Texture: Silt Loam)	0–5	19	55	0	0.45	7.0	13.5	0.02
	5–30	14	57	3	0.44	7.0	13.5	0.02
	30–60	14	52	0	0.42	7.0	13.2	0.01
	60–90	16	59	0	0.42	7.0	13.3	0.01
	90–120	12	61	0	0.41	7.0	13.3	0.01
Sherpur location 4 (Latitude/Longitude: 29.501 N/ 77.687 E, Texture: Sandy Loam)	0–5	12	31	0	0.33	7.5	16.2	0.02
	5–30	11	29	0	0.33	7.5	15.9	0.02
	30–60	10	25	0	0.32	7.5	15.9	0.02
	60–90	15	23	0	0.31	7.5	15.9	0.02
	90–120	17	27	0	0.31	7.4	16.0	0.02
Jharkheri location 5 (Latitude/Longitude: 29.370 N/ 77.190 E, Texture: Sandy Loam)	0–5	9	28	0	0.77	6.9	22.1	0.05
	5–30	11	25	0	0.75	6.9	22.1	0.05
	30–60	7	29	0	0.75	6.9	22.1	0.05
	60–90	10	24	1	0.75	6.8	21.8	0.04
	90–120	13	19	2	0.74	6.8	21.8	0.05
Raipur location 6 (Latitude/Longitude: 29.510 N/ 77.391 E, Texture: Loam)	0–5	25	32	0	0.62	8.1	18.9	0.03
	5–30	25	35	0	0.59	8.1	19.0	0.01
	30–60	22	38	0	0.59	8.0	18.9	0.01
	60–90	19	35	0	0.56	7.9	18.9	0.01
	90–120	15	34	0	0.56	7.9	18.8	0.02
Gangery location 7 (Latitude/Longitude: 29.310 N/ 77.220 E, Texture: Sandy Loam)	0–5	8	27	0	0.54	7.5	18.9	0.03
	5–30	6	24	0	0.54	7.5	19.0	0.01
	30–60	9	23	0	0.53	7.5	17.9	0.01
	60–90	5	27	1	0.54	7.5	17.9	0.01
	90–120	7	29	1	0.52	7.4	17.7	0.01
Jarouda location 8 (Latitude/Longitude: 29.390 N/ 77.679 E, Texture: Sandy Loam)	0–5	17	23	0	0.21	7.8	13.1	0.05
	5–30	15	20	0	0.21	7.8	12.7	0.05
	30–60	16	18	0	0.19	7.8	12.7	0.05
	60–90	12	15	2	0.17	7.6	12.8	0.03
	90–120	15	19	3	0.17	7.6	12.7	0.03
Sikanderpur location 9 (Latitude/Longitude: 29.263 N/77.657 E, Texture: Sandy Loam)	0–5	15	19	0	0.76	7.7	18.9	0.03
	5–30	16	15	0	0.75	7.7	19.0	0.03
	30–60	16	17	0	0.71	7.7	18.7	0.03
	60–90	14	21	0	0.71	7.7	18.9	0.01
	90–120	14	18	0	0.71	7.7	18.9	0.01
Gadanpur location 10 (Latitude/Longitude: 29.208 N/ 77.840 E, Texture: Silt Loam)	0–5	19	59	0	0.37	8.2	15.2	0.02
	5–30	21	57	0	0.35	8.2	15.2	0.02
	30–60	18	54	0	0.32	8.1	14.9	0.02
	60–90	20	56	0	0.32	8.0	14.9	0.01
	90–120	23	51	1	0.31	7.9	14.9	0.01
Lalpur location 11 (Latitude/Longitude: 29.293 N/ 78.068 E, Texture: Loam)	0–5	15	38	0	0.65	7.5	14.6	0.05
	5–30	18	37	0	0.65	7.5	14.6	0.05
	30–60	15	37	1	0.59	7.5	14.6	0.02
	60–90	19	36	0	0.55	7.4	14.1	0.02
	90–120	19	39	0	0.51	7.2	14.1	0.02
Kamalpur location 12 (Latitude/Longitude: 29.612 N/ 77.141 E, Texture: Silt Loam)	0–5	11	58	0	0.68	7.9	11.3	0.04
	5–30	13	57	0	0.68	7.9	11.2	0.03
	30–60	10	52	0	0.67	7.9	11.1	0.03
	60–90	14	54	0	0.66	7.8	11.1	0.03
	90–120	15	55	1	0.66	7.8	11.1	0.02

Meerut locations of Uttar Pradesh) by (Singh et al., 2010; Bhengra et al., 2016). There are twenty parameters of the genetic coefficient that need to be adjusted in the CANEGRO model for each variety. The values of the genetic coefficient of selected varieties are given in Table 2.

3.8. Crop management file

“The X build module” of the DSSAT generates crop management files. This module requires data of field operations (planting, harvesting, irrigation, fertilizer application, etc.) as specified by the users. In this

file, three planting dates of sugarcane crop, namely early-planting (P1), mid-planting (P2), and late-planting (P3) are taken. Various required field operations were performed for the sugarcane crop the schedule of the experiment are given in Table 3.

3.9. Model experiments for sugarcane yield assessment

The research work has been performed with the CANEGRO model under the given inputs of weather, soil, and crop genetic coefficients for a particular management practice defined in the experimental

Table 2

Parameters of genetic coefficients used in CANEGRO model.

S.No	Parameter/ code	Genetic coefficients of sugarcane varieties				
		CoS-767	CoSe- 95422	CoS-8436	CoSe-92423	CoSe-98231
1	PARCEmax	8.85	9.5	8.4	8.4	8.3
2	APFMX	0.88	0.88	0.83	0.88	0.88
3	STKPFMAX	0.65	0.65	0.55	0.6	0.6
4	SUCA	0.58	0.58	0.64	0.64	0.64
5	TBFT	25	25	28	26.8	27
6	Tthalfo	250	250	220	250	250
7	Tbase	16	16	16	16	16
8	LFMAX	12	12	10	12	12
9	MXLFAREA	360	370	345	360	370
10	MXLFARNO	15	16	14	16	16
11	PI1	89	110	95	100	90
12	PI2	176	200	180	220	170
13	PSWITCH	16	15	15	15	13
14	TTPLNTEM	340	360	340	360	350
15	TTRATNEM	203	203	203	203	203
16	CHUPIBASE	1740	1580	950	950	750
17	TT_POPGROWTH	610	580	610	600	550
18	MAX_POP	26	24	22	23	23.5
19	POPTT16	11.5	10	11	9.5	10.2
20	LG_AMBASE	220	220	220	220	220

Table 3

Field observations for sugarcane crop (2013–2014).

S. No.	Field Operation	Early Planting	Mid Planting	Late Planting
Field preparation				
1.	Tillage	21–01–2013	03–03–2013	17–04–2013
2.	Organic Amendments (Barnyard manure)	27–01–2013	09–03–2013	22–04–2013
Fertilizer application				
3.	N:P:K; (60:60:40)	30–01–2013	12–03–2013	24–04–2013
4.	N:P:K; (60:0:0)	17–03–2013	29–03–2013	07–06–2013
5.	N:P:K; (60:0:0)	05–04–2013	16–05–2013	13–07–2013
Planting				
6.	Horizontally planted stick (depth 10–15 cm)	02–02–2013	15–03–2013	28–04–2013
Emergence				
7.	After 40–45 days of planting	11–03–2013	25–04–2013	09–06–2013
Irrigation				
8.	1st irrigation (flood, mm)	13–03–2013	23–04–2013	02–06–2013
9.	2nd irrigation (flood, mm)	30–03–2013	11–05–2013	09–07–2013
10.	3rd irrigation (flood, mm)	21–04–2013	31–05–2013	04–09–2013
11.	4th irrigation (flood, mm)	09–05–2013	06–07–2013	23–09–2013
12.	5th irrigation (flood, mm)	01–06–2013	19–09–2013	07–10–2013
13.	6th irrigation (flood, mm)	09–07–2013	03–10–2013	29–10–2013
14.	7th irrigation (flood, mm)	14–09–2013	31–10–2013	21–02–2014
15.	8th irrigation (flood, mm)	05–10–2013	–	–
16.	9th irrigation (flood, mm)	27–10–2013	–	–
Weeding				
17.	Manually	As and when required		
Harvesting				
18.	Manually	03–02–2014	05–02–2014	08–03–2014

treatments. Proper selection of treatments can simulate the yield variability created due to the use of different management practices. These are related to the initial soil-water and inorganic nitrogen conditions, seed-bed preparation, planting geometries (planting date, plant population, and row spacing data), irrigation and water management

(irrigation dates, irrigation amount), organic residue applications, chemical applications, tillage operations, environmental modifications, and harvest management. Any change in these management can be introduced, and its interaction with all other yields affecting factors and final yield can be simulated.

3.10. Model experiments for climate change impacts on sugarcane yield assessment

For the study of climate change impact on sugarcane yield, a new WTH.DIR weather file for the sugarcane research farm sites (Chhapar location) was created in weatherman module of DSSAT. Daily weather parameters (minimum temperature, maximum temperature, solar radiation, rainfall) of past thirty five years from 1979 to 2013 was averaged to create a single weather file for the year 2013. This data were considered as baseline data to obtain a base yield. This research work has been performed with the CANEGRO model under the given inputs of weather, soil and crop genetic coefficients for a particular management practice defined in the experimental treatments. All packages of practices, soil parameters file and other input files were remain same other than weather parameter file to predict the effects of climate variables alone on sugarcane yield. Impact assessment of climate change in yield of sugarcane has been made putting different values of Tmax (°C), Tmin (°C), solar radiation (MJ/m²/day), and CO₂ levels (ppm) in the model. As the higher yielded variety CoSe-98231 and lower yielded variety CoS-767 have been taken for analysis so the base yield for CoSe-98231 and CoS-767 are obtained 845.6 (Qt/ha), 688.7 (Qt/ha) respectively. The planting date 15th March 2013 and harvesting date 5th February 2014 are used for this simulation.

4. Results

4.1. Models calibration and validation

In order to calibrate the CANEGRO model, field experiments were performed at twelve locations of Muzaffarnagar District under three different dates of planting i.e., 2nd February 2013 (early-planting), 15th March 2013 (mid-planting) and 28th April 2013 (late-planting). Data on growth, development, and yield were recorded for each planting date during field visits and created 180 treatment combinations using five sugarcane varieties at twelve different locations. Thereafter, the CANEGRO model was executed using actual crop management, soil, weather data, and genetic coefficient data. The CANEGRO simulated sugarcane

yield for the variety CoS-767, CoSe-95422, CoS-8436, CoSe-92423, and CoSe-98231 during the spring season (2013–2014) is presented in Table 4. To validate the performance of the model, input data required by the CANEGRO were provided for two consecutive years (2014–15 and 2015–16), and sugarcane district level yield data were used for validation. The twelve different locations sugarcane yield data were collected during field visits while the district level data were collected from the Directorate of Agriculture, Lucknow (Uttar Pradesh). The CANEGRO simulated variety-wise sugarcane yield for the entire district during spring season 2014–2015 and 2015–2016 are presented in Table 5. The generated models are calibrated and validated under the three different dates of planting.

4.2. Models accuracy assessment

To evaluate the performance of CANEGRO model, different statistical indicators, such as percent deviation, coefficient of determination (R^2), correlation coefficient (R), root mean square error (RMSE), mean absolute percent error (MAPE), and Willmott index of agreement (D) are used for the accuracy assessment of model results. The coefficient of determination (R^2) between observed and simulated sugarcane yield for twelve different locations of Muzaffarnagar District (2014–15 and 2015–16) is presented in Fig. 4, and the other parameters (R, RMSE, MAPE, and D) are shown in Fig. 5.

In Fig. 4, R^2 values between observed and predicted yield for the year 2014–15 is 0.60, 0.66, and 0.62 for three simulation models, i.e., early, mid, and late-planting sugarcane, respectively. Similar, R^2 values were found for the year 2015–16 as 0.62 for early-planting model, 0.70 for mid-planting model, and 0.57 for late-planting model. Correlation represents the relationship between two quantitative variables and the values of correlation coefficient (R) varies from -1 to $+1$. In this study, R values (Fig. 5a) between observed and predicted sugarcane yield were found as 0.77, 0.81, 0.78 (year 2014–15), and as 0.78, 0.83, 0.75 (year 2015–16) for early, mid, and late-planting models, respectively.

In addition, the performance of the models was also evaluated by RMSE and MAPE parameters.

RMSE is the square root of the variance of the residuals that indicates the absolute fit of the model to the data how close the observed data points are to the model's predicted values. The Mean absolute percentage error (MAPE) is one of the most commonly used KPIs (key performance indicator) to measure forecast accuracy. It is the sum of the individual absolute errors divided by the demand (each period separately) that measures the deviation (in terms of %) from the actual data. The RMSE values are 10.4 q ha^{-1} for early-planting, 8.37 q ha^{-1} for mid-planting, and 12.29 q ha^{-1} for late-planting for the year 2014–15, while these were 22.21 q ha^{-1} for early-planting, 10.7 q ha^{-1} for mid-planting, and 12.64 q ha^{-1} for late-planting for the year 2015–16 (Fig. 5b). In this study, very low MAPE values were observed for all three planting models. The ranges varies between 1.10 % and 1.90 % and 1.30–3.10 % for the year 2014–15 and 2015–16, respectively (Fig. 5c).

Willmott index of agreement (D) is a measure of the degree of model prediction error, and its value ranges between 0 (indicates no agreement) to 1 (indicates a perfect match). This represents the ratio of the

mean square error and the potential error, and can also detect additive and proportional differences in the observed and simulated means and variances. It is seen in Fig. 5(d) that the values of D ranges between 0.81 and 0.88 for the year 2014–15, and between 0.69 and 0.91 for the year 2015–16, respectively. This index value is highest for the mid-planting models (0.88 for 2014–15; 0.91 for 2015–16) as compared to early and late planting models which indicates that the CANEGRO model can be used for sugarcane yield simulation with relatively good accuracy.

The sugarcane yield simulation by mid-planting model was found to be better, presenting the highest R (0.81, 0.83), and D (0.88, 0.91), and the lowest errors (RMSE = 8.37 q ha^{-1} , 10.70 q ha^{-1} and MAPE = 1.10 %, 1.30 %) for the years 2014–15 and 2015–16, respectively, as compared to the other two models. The district-level predicted sugarcane yield by CANEGRO model shows the percent error ranges from $+2.80$ to $+31.64$. The model gave percent error from $+8.62$ to $+9.60$ for early-planting, $+20.13$ to $+31.64$ for late-planting, and $+2.80$ to $+5.20$ for mid-planting. The yield estimation for the mid-planting model thus proved superior over early and late-planting models (Table 6). The simulated sugarcane yield of different varieties during 2013–14, 2014–15, and 2015–16 are presented in Fig. 6.

5. Discussion

5.1. Climate change impact on yield

The climate scenario simulated for maximum temperature (± 1 to $\pm 3 \text{ }^\circ\text{C}$), minimum temperature (± 1 to $\pm 3 \text{ }^\circ\text{C}$), solar radiation (± 1 to $\pm 3 \text{ MJ/m}^2/\text{day}$), and (500 ppm, 720 ppm) against present CO_2 concentration of 380 ppm. Looking for the complexity in climatic pattern, the yield was simulated due to change in individual parameters (Table 7) and in combination with other parameters (Table 8).

5.2. Effect of Tmax on yield

The CANEGRO model simulated yield under incremental values of Tmax in the range of $(+1$ to $+3 \text{ }^\circ\text{C})$ which shows gradual decrement in the yield range of 677.6 – 664.0 Qt ha^{-1} (-0.09 , -0.33 and -2.09 % of base yield) for variety CoS-767. We observed that the response for incremental values of Tmax is similar for variety CoSe-98231, except that there is a higher reduction in yield as compared to variety CoS-767. For variety CoSe-98231, the yield is decreasing in the range of 824.0 – 798.2 Qt ha^{-1} (-1.03 , -2.34 and -4.13 % of base yield). Similarly, in case of gradual downscale of Tmax in the range of -1 to $-3 \text{ }^\circ\text{C}$, we have noticed an incremental pattern in yield ranging from 682.4 to 692.3 Qt ha^{-1} ($+0.62$, $+1.16$ and $+2.08$ % of base yield) for variety CoS-767. In the range of -1 to $-3 \text{ }^\circ\text{C}$, The variety CoSe-98231 yield is also increasing in the range of 843.6 – 857.1 Qt ha^{-1} ($+1.32$, $+2.07$ and $+2.94$ % of base yield). The results show that the variety CoSe-98231 is more sensitive to the maximum temperature change as compared to variety CoS-767.

5.3. Effect of Tmin on yield

The increment of Tmin in the range of $(+1$ to $+3 \text{ }^\circ\text{C})$ shows gradual decrement in the yield range of 670.2 – 642.2 Qt ha^{-1} (-1.18 , -2.64 and -5.31 % of base yield) for variety CoS-767. The same results were obtained for the variety CoSe-98231. The yield decreases in the range of 822.5 – 793.2 Qt ha^{-1} (-1.21 , -2.68 and -4.73 % of base yield). We have noticed that the yield decrement is maximum at the $+3 \text{ }^\circ\text{C}$ for both the variety. In case of downscaling of Tmin from -1 to $-2 \text{ }^\circ\text{C}$, the sugarcane variety CoS-767 yield increases from 680.3 to 683.3 Qt ha^{-1} ($+0.31$ to $+0.75$ % of base yield) while it decreased by -1.64 % of base yield (678.2 Qt ha^{-1}) at $-3 \text{ }^\circ\text{C}$. The CoSe-98231 variety yield is gradually increasing in the range of 844.6 – 857.6 Qt ha^{-1} ($+1.44$, $+2.16$ and $+3.00$ % of base yield) for the scale of Tmin -1 to $-3 \text{ }^\circ\text{C}$.

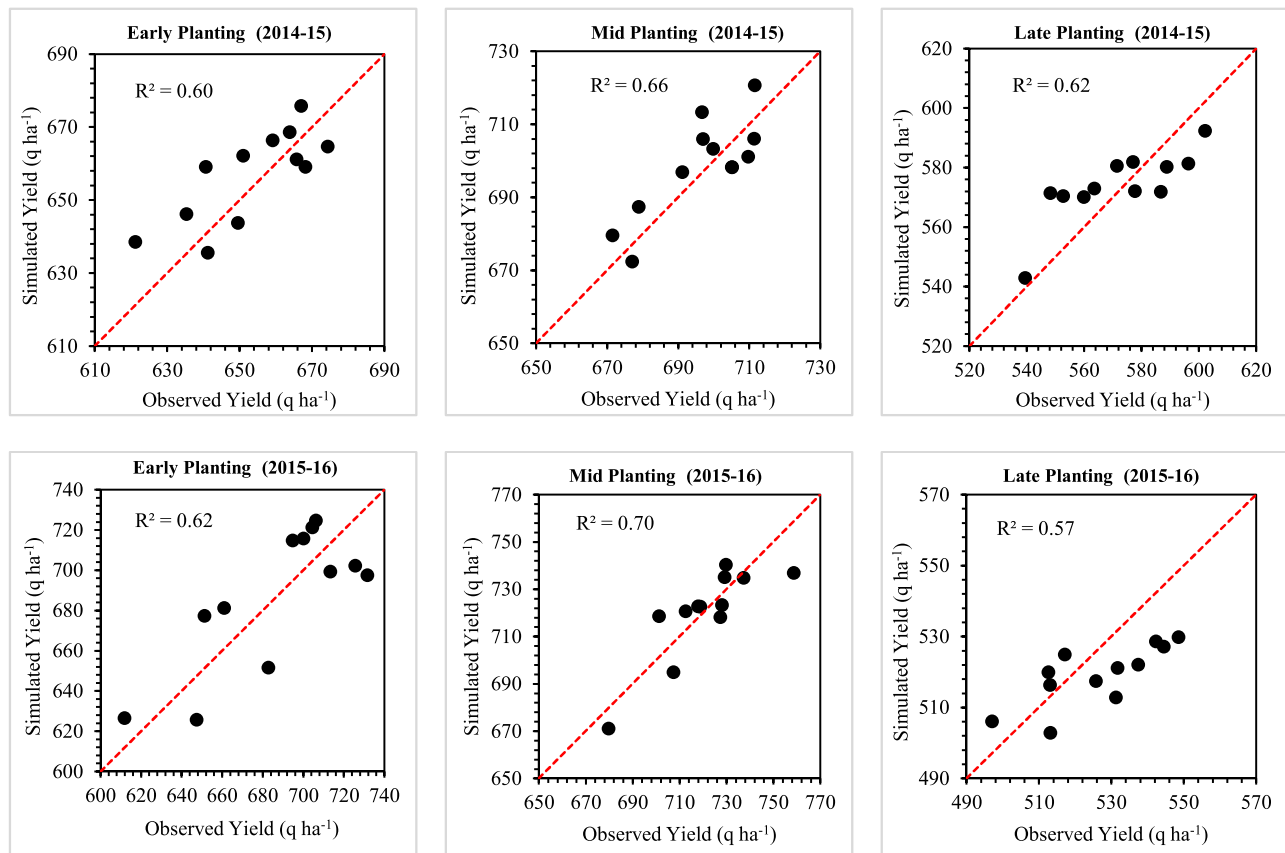
Table 4
CANEGRO simulated variety-wise sugarcane yield for Muzaffarnagar district (2013–14).

S. No.	Variety	Sugarcane Yield (q ha^{-1})		
		Early planting	Mid planting	Late planting
1	CoS-767	624.11	687.80	504.47
2	CoSe-95422	720.84	765.08	607.31
3	CoS-8436	635.33	650.20	527.50
4	CoSe-92423	698.80	718.45	592.95
5	CoSe-98231	786.20	794.33	666.95
	Average	693.05	723.17	579.83

Table 5

CANEGRO simulated variety-wise sugarcane yield for Muzaffarnagar district (2014–15 and 2015–16).

S. No.	Variety	2014–2015 Yield (q ha ⁻¹)			2015–2016 Yield (q ha ⁻¹)		
		Early Planting	Mid Planting	Late Planting	Early Planting	Mid Planting	Late Planting
1	CoS-767	559.71	630.67	446.86	600.46	658.50	459.56
2	CoSe-95422	685.55	755.18	581.50	702.99	750.28	540.58
3	CoS-8436	601.75	619.12	533.22	628.48	638.71	468.73
4	CoSe-92423	674.47	698.65	614.68	702.02	731.42	526.69
5	CoSe-98231	762.32	789.56	694.0	798.20	820.63	599.73
	Average	656.76	698.63	574.05	686.43	719.91	519.06

**Fig. 4.** Coefficient of determination between observed and simulated sugarcane yield at twelve different locations of Muzaffarnagar district for the year 2014–15 and 2015–16.

5.4. Effect of solar radiation on yield

For both the varieties, gradual increment of solar radiation from 1 to 3 MJm⁻²day⁻¹ showed a yield increment (+8.82, +16.48 and +23.49 % of base yield for CoS-767) and (+7.23, +13.86 and +19.66 % of base yield for CoSe-98231). In case of decrement of solar radiation from (–1 to –3 MJm⁻²day⁻¹) reverse trend was noticed. The maximum increment in yield was reported (+23.49 % of base yield) for variety CoS-767 at 3 MJm⁻²day⁻¹ whereas the maximum reduction in yield was noticed (–27.13 % of base yield) for variety CoS-767 with reduction of solar radiation by 3 MJm⁻²day⁻¹. The results indicate that both the sugarcane varieties are very sensitive to the solar radiation effect, but the variety CoS-767 is more sensible as compared to CoSe-98231.

5.5. Effect of CO₂ concentration on yield

Increased levels of CO₂ directly affected crop species, which changes the crop physical structures and in turn affects crop growth and yield, water use efficiency (Ritchie and Otter, 1985) and susceptibility to

insect pests and diseases (Mitchell et al., 2003; PaHerson, 1993). The impact of increasing CO₂ concentration in the level of 720 ppm and 500 ppm, as their base value of 380 ppm showed a gradual increment in sugarcane yield for both the variety. In the high-emission A2 scenario for CO₂ concentration of 720 ppm, variety CoS-767 and CoSe-98231 yield were increased by + 5.69 % and 6.02 % of base yield, respectively. In case of B2 (Low emission scenario), CO₂ concentration of 550 ppm, the yield is incrementing by + 3.34 for variety CoS-767 and + 3.39 for variety CoSe-98231.

5.6. Combined effect of Tmax and Tmin on yield

Both the Tmax and Tmin temperatures are changed in the range of (± 1 to ± 3 °C) and their simulated yield levels are compared with base yield (Table 5). The CANEGRO model yield of sugarcane under incremental units of Tmax and Tmin (+1 to +3 °C) showed a gradual decrement in yield ranging from 667.5 to 606.6 Qt ha⁻¹ (–1.58, –5.06 and –10.56 % of base yield) for variety CoS-767. The similar decreasing yield pattern was noticed for the variety CoSe-98231 having ranges from

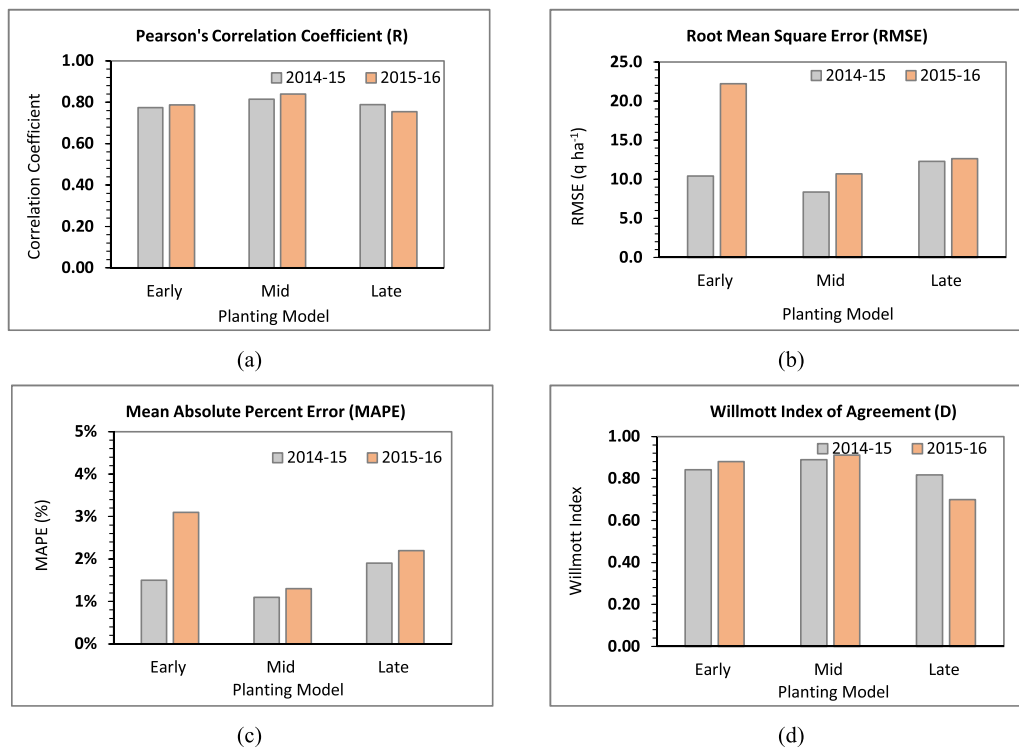


Fig. 5. Statistical parameters (R, RMSE, MAPE, D) for models accuracy assessment.

Table 6

The predicted sugarcane yield and percent deviation error by CANEGRO model.

Year	Model	CANEGRO predicted sugarcane yield ($q\ ha^{-1}$)	Error (%)
2013 (Calibration)	Early planting model	693.05	+4.98
	Mid planting model	723.17	+0.85
	Late planting model	579.83	+20.51
2014 (Validation)	Early planting model	656.76	+8.62
	Mid planting model	698.63	+2.80
	Late planting model	574.05	+20.13
2015 (Validation)	Early planting model	686.43	+9.60
	Mid planting model	719.91	+5.20
	Late planting model	519.06	+31.64

811.8 to 726.7 $Qt\ ha^{-1}$ (−2.50, −6.73 and −12.72 % of base yield). Other combinations did not show any significant trend. The drastically yield reduction was observed at −3 °C for variety CoS-767.

5.7. Combined effect of Tmax, Tmin and solar radiation on yield

Maximum temperature (Tmax), minimum temperatures (Tmin) and solar radiation are altered in the range of (± 1 to ± 3 °C) and their yield was compared with base yield (Table 5). The maximum increment of yield was reported at +2 °C for both the variety. The variety CoS-767 and CoSe-98231 yield were increased by +9.75 and +4.46 % of base yield, respectively. The maximum yield reduction was noticed at −3 °C. The yield is decreased by −39.47 % (variety CoS-767) and −31.10 % (Variety CoSe-98231) of base yield. The yield is drastically decreasing at −2 °C and −3 °C.

6. Conclusions

The DSSAT-CANEGRO was able to simulate sugarcane yield fairly realistically when compared with the observed yield. These models were successfully calibrated and validated for the Muzaffarnagar District under three different dates of planting. Sugarcane yield simulation mid-planting model presented the highest R (0.81, 0.83), and D (0.88, 0.91), and the lowest errors (RMSE = 8.37 $q\ ha^{-1}$, 10.70 $q\ ha^{-1}$ and MAPE = 1.10 %, 1.30 %) for the years 2014–15 and 2015–16 as compared to the other two models. The mid-planted sugarcane district-level yields percent error ranged from +2.80 to +5.20 which is the minimum among all the three planting models. The results revealed that the mid-planting model (3rd week of March) gives higher sugarcane yield as compared to the early and late planting models. Sugarcane requires about 25–32 °C for good germination therefore, early (before 2 FN of February) and late-planting (after 1 FN of April) may be the cause of yield loss due to low and high temperatures, respectively.

The genetic potential of a crop variety also plays an important role in determining the yield. Some varieties have a high yield potential while others have lower. The CANEGRO model simulated the yield for the five sugarcane varieties, namely CoS-767, CoSe-95422, CoS-8436, CoSe-92423 and CoSe-98231. Results showed that the variety CoSe-98231 produces higher yields, whereas variety CoS-767 gives the lowest yield for all three planting dates. Therefore, varieties CoSe-98231, CoSe-95422, CoSe-92423 are recommended for sugarcane growers and sugar industries for higher cane yield.

Sensitivity analysis has been also successfully performed for Chhapar site location to understand the effect of climate variability on sugarcane yield. CANEGRO model was simulated for single weather parameters (solar radiation, Tmax, Tmin) as well as combinations of these parameters. This study indicates that the CANEGRO model simulated yield under incremental values of Tmax and Tmin in the range of (+1 to +3 °C) shows gradual decrement in the yield ranges for both the sugarcane varieties while, gradual increment of solar radiation from 1 to 3 $MJ/m^2/day$ showed a yield increment (+8.82, +16.48 and +23.49 % of base yield for CoS-767) and (+7.23, +13.86 and +19.66 % of base yield for

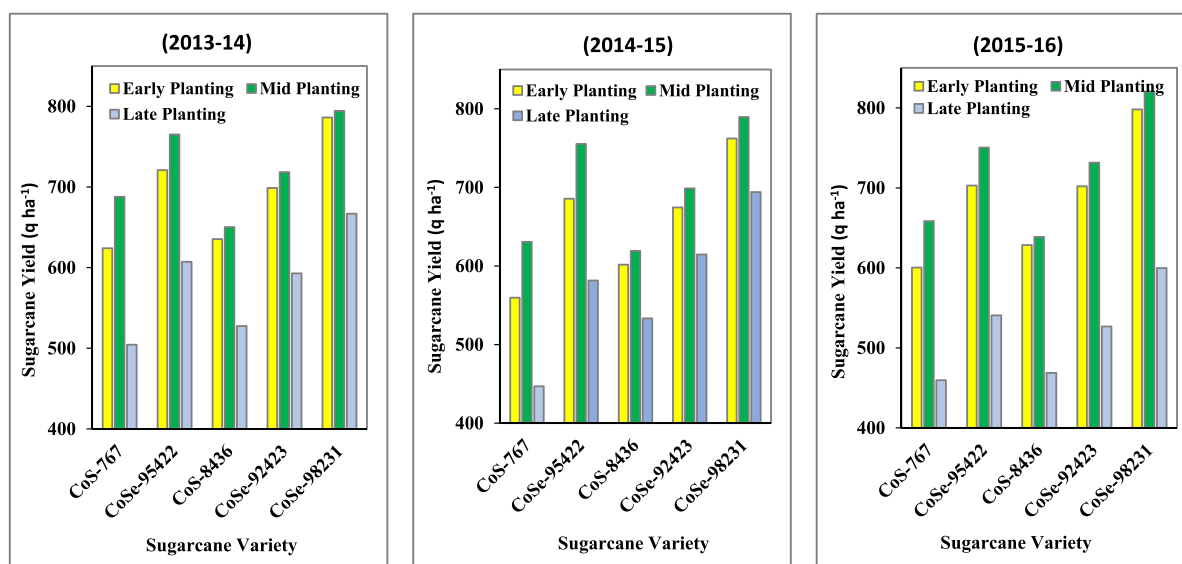


Fig. 6. Simulated yield for five sugarcane varieties for the year 2013–14, 2014–15, and 2015–16.

Table 7

Sensitivity analysis of CANEGRO model for various experimental values of Tmin, Tmax, solar radiation and CO₂ concentration on yield.

Parameters	CoS-767		CoSe-98231	
	Simulated Yield (Qt/ha)	% Change from base yield	Simulated Yield (Qt/ha)	% Change from base yield
Tmax (°C)				
+ 3°	664.0	-2.09	798.2	-4.13
+ 2°	675.9	-0.33	813.1	-2.34
+ 1°	677.6	-0.09	824.0	-1.03
-1°	682.4	+0.62	843.6	+1.32
-2°	686.1	+1.16	849.8	+2.07
-3°	692.3	+2.08	857.1	+2.94
Tmin (°C)				
+ 3°	642.2	-5.31	793.2	-4.73
+ 2°	660.3	-2.64	810.3	-2.68
+ 1°	670.2	-1.18	822.5	-1.21
-1°	680.3	+0.31	844.6	+1.44
-2°	683.3	+0.75	850.6	+2.16
-3°	678.2	-1.64	857.6	+3.00
Solar radiation (MJ/m ² /day)				
+ 3	837.5	+23.49	996.3	+19.66
+ 2	790	+16.48	948	+13.86
+ 1	738	+8.82	892.8	+7.23
-1	616.1	-9.16	773.6	-7.09
-2	554.8	-18.20	710.6	-14.65
-3	494.2	-27.13	642.6	-22.82
CO ₂ concentration (Base value 380 ppm)				
720 ppm (A2)	716.8	+5.69	882.8	+6.02
500 ppm (B2)	700.9	+3.34	860.9	+3.39

CoSe-98231). The study highlighted that the generated sugarcane yield model is sensitive to the solar radiation, maximum temperature, minimum temperature and CO₂ concentration levels.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

Table 8

Sensitivity analysis of CANEGRO model for the combined effect of Tmax, Tmin and solar radiation on yield.

Parameters	CoS-767		CoSe-98231	
	Simulated Yield (Qt/ha)	% Change from base yield	Simulated Yield (Qt/ha)	% Change from base yield
Tmax and Tmin (°C)				
+3°	606.6	-10.56	726.7	-12.72
+2°	643.9	-5.06	776.6	-6.73
+1°	667.5	-1.58	811.8	-2.50
-1°	679	0.12	849.9	2.08
-2°	639.3	-5.74	840.2	0.91
-3°	570.9	-15.82	765.2	-8.10
Tmax (°C), Tmin (°C) and solar radiation (MJ/m ² /day)				
+3°	725.2	+6.93	850.6	+2.16
+2°	744.3	+9.75	869.7	+4.46
+1°	725.4	+6.96	865.1	+3.90
-1°	617.6	-8.94	785.3	-5.68
-2°	524.2	-22.71	706.5	-15.15
-3°	410.5	-39.47	573.7	-31.10

the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Compliance with ethical standards

The authors declare compliance with ethical standards.

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